

Lecture 40 — SVD, low-rank approximation, and capstone launch

Fri Dec 4

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: selected [Strang/MIT SVD slides](#), [Lab 6](#), [Quiz 10](#), capstone launch, Lachniet Ch. 5 and Ch. 7. Related objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Interpret $A = U\Sigma V^T$ conceptually.
2. Construct a rank- k approximation computationally.
3. Compare singular-value decay for an image/frame and a frame-difference matrix.
4. Encode three frames either independently or as F_1, D_2, D_3 .
5. Compare reconstruction error and storage for the two approaches.
6. Formulate a capstone question involving accuracy, storage, runtime, and resource use.